# Adam in a Fixed Hessian Eigenbasis

AISC5020: Module 5 Method and Verification Note

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**Abstract**

This module studies a controlled intervention for coordinate dependence: apply Adam in a fixed orthonormal eigenbasis of a positive-definite quadratic, then map each update back to the original coordinates. We prove exact trajectory correspondence with Adam on the diagonal problem, including positive numerical stabilization and bias correction. We distinguish the common-basis statement from the stronger, generally invalid claim that arbitrary eigenbases give the same trajectory when eigenvalues are repeated. The implementation separates basis preparation, residual diagnostics, optimization, and evaluation costs. This note documents Module 5 only; it does not report a tuned comparison of all team methods or claim completion of the group's main experiment.

## 1 Question and controlled intervention

Consider

$$f(w) = \tfrac{1}{2} w^\top H w, \qquad H = V D V^\top, \qquad V^\top V = I, \qquad D = \operatorname{diag}(\lambda_1, \dots, \lambda_p), \quad \lambda_i > 0. \tag{1}$$

The eigenvalues determine the quadratic's spectral conditioning. They do not specify the axes along which Adam accumulates squared gradient components. Ordinary Adam is therefore an appropriate object for a coordinate experiment, but changing coordinates and observing different convergence curves alone does not identify which operation causes the difference.

Our intervention fixes the coordinates used for moment accumulation. With $z = V^\top w$, the same objective becomes $\phi(z) = \frac{1}{2} z^\top D z$. At update $t$, let

$$g_t = H w_{t-1}, \qquad h_t = V^\top g_t. \tag{2}$$

We apply the bias-corrected Adam recurrence [**?**] to $h_t$:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) h_t, \qquad v_t = \beta_2 v_{t-1} + (1-\beta_2)(h_t \odot h_t), \tag{3}$$

$$\widehat{m}_t = \frac{m_t}{1-\beta_1^t}, \qquad \widehat{v}_t = \frac{v_t}{1-\beta_2^t}, \tag{4}$$

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon \mathbf{1}}, \qquad w_t = w_{t-1} - \eta_t V d_t. \tag{5}$$

The square, square root, and quotient of equal-length vectors are componentwise; $V d_t$ is a matrix–vector product. We use $m_0 = v_0 = 0$, $\beta_1, \beta_2 \in [0, 1)$, a common scalar $\epsilon > 0$ outside the square root, and shared step sizes $\eta_t > 0$. The moments live in eigen-coordinates even though $w_t$ is stored in ambient coordinates. Transforming only the final direction while retaining ambient second moments would define a different method.

## 2 Exact correspondence and its scope

**Proposition 1** (P4: correspondence in a specified common basis)**.** *Run* (**??**)–(**??**) *on* $H = VDV^\top$ *from* $w_0 = Vz_0$*. Run ordinary Adam on* $\phi(z) = \frac{1}{2}z^\top Dz$ *from* $z_0$*, with the same parameters, step sizes, zero initial moments, and update indexing. In exact arithmetic, for every update* $t$*,*

$$w_t = Vz_t, \qquad f(w_t) = \phi(z_t). \tag{6}$$

*The result allows repeated eigenvalues and positive* $\epsilon$*.*

*Proof.* The initial points correspond by construction. Suppose $w_{t-1} = Vz_{t-1}$ and the two pairs of moment vectors agree. Orthogonality and (**??**) give

$$h_t = V^\top HVz_{t-1} = Dz_{t-1} = \nabla\phi(z_{t-1}).$$

Thus (**??**) produces identical first and second moments in the two runs. Their scalar bias corrections and positive scalar stabilizers also agree, so their directions $d_t$ agree, including when a gradient component is zero. Consequently,

$$w_t = Vz_{t-1} - \eta_t Vd_t = V(z_{t-1} - \eta_t d_t) = Vz_t.$$

Finally, $f(Vz_t) = \frac{1}{2}z_t^\top V^\top HVz_t = \phi(z_t)$. Induction proves both claims. □

This is a trajectory identity, not a convergence-rate theorem. It gives no guarantee that the chosen step size converges, that Adam outperforms GD, or that the additional basis information is inexpensive. It also requires a fixed basis: changing the basis during a run without consistently reinterpreting the moment history invalidates the argument.

**Lemma 1** (Signed-permutation ambiguity)**.** *Let* $R$ *be a signed permutation matrix and replace* $V$ *by* $V' = VR$*. With corresponding coordinate initialization, the two eigenbasis-Adam runs have the same ambient iterates in exact arithmetic.*

*Proof.* Write $P = |R|$, an unsigned permutation matrix. Coordinate gradients and first moments transform as $h'_t = R^\top h_t$ and $m'_t = R^\top m_t$, whereas second moments transform as $v'_t = P^\top v_t$. This follows inductively because $(R^\top h) \odot (R^\top h) = P^\top(h \odot h)$. Bias correction commutes with both transformations. Since $P^\top \mathbf{1} = \mathbf{1}$ and the denominator is permuted without a sign, the normalized direction obeys $d'_t = R^\top d_t$. Therefore $V'd'_t = VRR^\top d_t = Vd_t$, so the physical updates coincide. □

For a simple spectrum, every exact orthonormal eigenbasis of the same $H$ differs by such a signed permutation. A basis recovered from $H$ alone therefore has the same idealized physical trajectory; aligning its signs to the generating basis is unnecessary. Numerically close eigenvalues require more care because their eigenvectors can be sensitive to small perturbations. In particular, mathematically distinct eigenvalues do not by themselves guarantee numerically resolvable eigenvectors or agreement of long floating-point trajectories.

For repeated eigenvalues, an eigensolver may choose an arbitrary orthogonal rotation within a repeated eigenspace. Adam's componentwise square does not commute with a general rotation, so Lemma **??** does not apply. For example, let $H = I_2$, $w_0 = (1, 0)^\top$, and let $R$ be a rotation through $\pi/4$. Both $I_2$ and $R$ are valid eigenbases. At the first update their ambient Adam directions are, respectively,

$$\frac{(1,0)^\top}{1+\epsilon} \quad \text{and} \quad \frac{(1,0)^\top}{1/\sqrt{2}+\epsilon},$$

which differ for every $\epsilon > 0$. Proposition **??** remains valid when both runs use one specified common basis; arbitrary independent eigenbasis recovery is a different claim.

## 3 Implementation and verification protocol

Basis preparation is explicit and occurs before optimization. The implementation supports three sources. An *oracle* basis is the generating $V$ and isolates the algebraic mechanism. An *eigh* basis is obtained from the Hessian alone using a symmetric eigensolver; the generating rotation is not used to align its columns. A *supplied* basis permits controlled sign, permutation, and degenerate-subspace tests. The selected basis is copied, validated, and held fixed, and a prepared object may be reused only for a matching Hessian.

We distinguish basis quality from trajectory agreement. For a candidate basis $B = [b_1, \dots, b_p]$ and $\widetilde{D} = \operatorname{diag}(b_1^\top H b_1, \dots, b_p^\top H b_p)$, report

$$r_{\mathrm{orth}} = \frac{\left\| B^\top B - I \right\|_F}{\sqrt{p}}, \qquad r_{\mathrm{diag}} = \frac{\left\| HB - B\widetilde{D} \right\|_F}{\|H\|_F}. \tag{7}$$

Positive eigenvalues, finite arrays, dimensions, symmetry, and residual tolerances are checked separately. Numerical eigenvalue clusters are recorded as a warning about basis ambiguity; a tolerance-based cluster is not proof of an exactly repeated eigenvalue. The diagnostics also compare the minimum eigengap with a conservative Hessian-scale allowance based on residuals, roundoff, and the accepted validation tolerance. Stored eigenvalues may be distinct while the eigenvector-uncertainty flag is true. This indicator is not a certified forward-error bound for a particular eigensolver.

For paired trajectories, retain both the ambient discrepancy

$$e_t = \frac{\|w_t - V z_t\|_2}{\max(1, \|w_0\|_2)} \tag{8}$$

and the absolute loss difference. A fixed initial scale avoids a misleading relative error whose denominator vanishes near the optimizer. Residuals in (**??**) answer whether $B$ approximately diagonalizes $H$; (**??**) answers a different, dynamical question. Neither should be reported as the other.

The verification suite covers the following distinct claims:

1. Compare every recorded iterate for a shared generating basis, with both distinct and repeated spectra.
2. Recover a basis from $H$ alone for a distinct spectrum and retain the eigensolver's signs and ordering.
3. Supply a signed and permuted basis to check Lemma **??**; separately supply an admissible rotation within a repeated eigenspace to expose the limitation.
4. Reject invalid bases and basis reuse with a different Hessian, and count preparation and transformation work explicitly.

Each numerical record should preserve the spectrum, initial point, seed, parameters, horizon, code version, and error definition. Module verification uses fixed settings to test the mechanism. It does not replace the team's independent tuning and statistical protocol.

### 3.1 Observed structural verification

The saved run of 15 October 2026 (08:36:01 UTC) uses seed 502015, 256 updates, $\eta = 0.01$, $(\beta_1, \beta_2) = (0.9, 0.999)$, and $\epsilon = 10^{-8}$. There are 84 two-dimensional cases, 24 cases with $p = 20$,

16 with $p = 100$, and two declared scope controls, for 126 cases in total. The main spectra use $\kappa \in \{10, 10^2, 10^3, 10^4\}$; the two-dimensional angles range from $0^\circ$ to $90^\circ$ in $15^\circ$ increments. Each case has one diagonal reference and three runs on the rotated problem: ordinary Adam, oracle-basis Adam, and recovered-basis Adam. Thus 504 trajectories are saved, including 378 rotated-problem trajectories.

All 250 required correspondence checks passed the preset normalized position tolerance $5 \times 10^{-8}$; Table **??** reports the observed maxima. The recovered-basis requirement applies to the 124 separated-spectrum cases, whereas the oracle common-basis requirement includes both controls. These settings test an identity rather than tune the methods for minimum loss.

Table 1: Maximum normalized path discrepancy over required checks.

| Method and scope | Cases | Maximum $e_t$ |
|---|---|---|
| Oracle basis, all cases | 126 | $1.535 \times 10^{-12}$ |
| Recovered basis, separated spectra | 124 | $6.486 \times 10^{-13}$ |

The controls make the scope visible rather than silently omitting difficult cases. For the repeated identity problem, the recovered-basis discrepancy is 0.1819, while the oracle common-basis discrepancy is $2.483 \times 10^{-16}$. For the near-repeated spectrum $(10^{-4}, 10^{-4} + 10^{-11}, 1)$, the recovered-basis discrepancy is $5.637 \times 10^{-7}$ despite a diagonalization residual $4.885 \times 10^{-16}$. Its stored eigenvalues are distinct, but the numerical uncertainty flag is true. The oracle discrepancy remains $1.535 \times 10^{-12}$. This illustrates why a small backward residual and distinct stored eigenvalues cannot alone establish forward trajectory agreement.

A separate checker recomputed all 504 saved trajectories from the saved problems using explicit Adam recurrences, without the production Adam update helper. It also independently regenerated the seeded case schedule and initial points, recomputed losses, summary metrics, basis residuals and timing medians, and checked operation counts, all 45 timing rows, and 19 source/configuration/script hashes. Its saved verification status is passed. The environment was Python 3.12.14, NumPy 2.5.4, and Windows 11, with BLAS thread environment variables set to one.

## 4 Cost accounting and numerical interpretation

For dense $p \times p$ matrices, recovering all eigenvectors costs $O(p^3)$ arithmetic and $O(p^2)$ storage. Validation also contains dense matrix–matrix products and belongs to setup. At each optimization step, $V^\top g_t$ and $V d_t$ add two dense matrix–vector products, each costing $O(p^2)$, as well as $O(p)$ moment operations. The common quadratic gradient $Hw$ already costs $O(p^2)$. Computing a loss for monitoring is additional evaluation work and must not be silently counted as a free gradient call.

Our accounting separates basis setup, the full evaluation loop, and total elapsed time. The evaluation loop includes gradients, coordinate transforms, losses, checks, and trace bookkeeping; operation counts distinguish these activities, but the implementation does not time each one separately. Counts are more portable than timings on small matrices. Reusing a prepared basis avoids a second eigendecomposition but does not erase the first preparation cost; original setup and reuse metadata must remain visible. Likewise, an oracle basis removes recovery from the timed optimization loop by assumption, not by providing a generally free algorithm.

Table **??** gives local medians from five repetitions per dimension and method, each with 256 updates. One untimed warm-up run per dimension and method is excluded from the five measured repetitions. These are descriptive timings, not evidence of a practical speedup. Setup

and total medians are calculated separately, so adding component medians need not reproduce a total median.

Table 2: Local median time in milliseconds. Basis-method cells show setup / total; ordinary Adam has no basis setup.

| $p$ | Ordinary total | Oracle setup / total | Recovered setup / total |
|---|---|---|---|
| 2 | 12.508 | 0.304 / 14.995 | 0.248 / 11.474 |
| 20 | 11.635 | 0.435 / 13.160 | 0.468 / 12.723 |
| 100 | 11.281 | 1.852 / 18.791 | 2.341 / 15.621 |

The proof uses exact arithmetic, while implementations repeatedly multiply floating-point matrices and accumulate moment histories. Small perturbations can be amplified by normalization, especially when gradients and loss are tiny. A long-horizon discrepancy after near-zero error is not a logical counterexample to Proposition **??**. It is still relevant empirical information: report its onset, magnitude, objective scale, and settings rather than hiding it or treating every late difference as a failed eigensolver.

Conversely, a small decomposition residual alone does not establish a uniformly small trajectory discrepancy for an arbitrary horizon. This report makes no forward-error bound or stability theorem of that form. Interpretation should combine algebra, independent implementation checks, basis residuals, and the observed trajectories.

The reproducible artifacts are `outputs/part5/summary.json`, `correspondence.csv`, `basis_diagnostics.json`, `timing.csv`, the saved raw trajectories, and `artifact_verification.json`. The same output directory contains `correspondence.pdf` and `cost.pdf`; this standalone source has no external figure dependency.

## 5 Integration and conclusion

The intervention gives the group a precise mechanism test. If ordinary Adam changes under a paired rotation while fixed-eigenbasis Adam tracks the diagonal reference over the same regime, the result supports the role of coordinatewise moment accumulation. That conditional interpretation must be based on actual paired outputs; it does not follow from a planned figure or from the theorem alone. Learning-rate selection, rotation sampling, uncertainty estimates, and comparison with preconditioning remain the responsibility of the corresponding team modules.

For integration, Module 5 contributes a fixed-basis implementation, a proof with explicit assumptions, diagnostics that separate recovery from dynamics, and cost metadata. The exact claim is deliberately limited: a specified common eigenbasis restores correspondence with diagonal-coordinate Adam; a simple spectrum makes signed-permutation ambiguity harmless; repeated eigenspaces retain a genuine coordinate choice. These distinctions allow a stronger mechanistic explanation without claiming universal rotation invariance or an unmeasured practical speedup.

## References

[1] D. P. Kingma and J. Ba. Adam: A Method for Stochastic Optimization. *International Conference on Learning Representations*, 2015. `https://arxiv.org/abs/1412.6980`.